Problem Set - 10

Kutay Kulbak

January 8, 2025

1 Mechanical Waves

1.1 Nice Reads

- Kevin Zhou Waves Handout

To be read **after** solving the problems:

- About the physics behind musical instruments.
- Lies!
- Why do drums sound like shit?

1.2 Waves on Rotating Hoop

A uniform circular hoop of string is rotating clockwise in the absence of gravity. The tangential speed is v. Find the speed of waves on this string. (Kevin Zhou, Waves I, P3)

1.3 Gravitational Waves

The power radiated in gravitational waves by an orbiting binary system is given by

$$P(r, m_1, m_2) = \frac{32}{5}\frac{G^4}{c^5}\frac{(m_1 m_2)^2(m_1 + m_2)}{r^5}$$

where r is the distance between the centers of the two orbiting masses m_1 and m_2. It is known that the most compact object is a black hole. The size of a black hole is defined by its Schwarzschild radius $r_s = \frac{2Gm}{c^2}$, where m is the mass of the black hole. (NBPhO 2017, P4)

(i) Estimate the upper limit of the power that can ever be emitted in gravitational waves by an orbiting binary system.

The gravitational wave detectors on Earth function by measuring the so called gravity wave strain $\varepsilon(t)$ over time, which characterizes the deformation of spacetime. Data processing then yields the maximum strain ε and its corresponding wave frequency f. With the help of a theoretical spacetime model, the energy density u associated with the wave can then be determined. We will use the analogy of linear elasticity to examine this model.

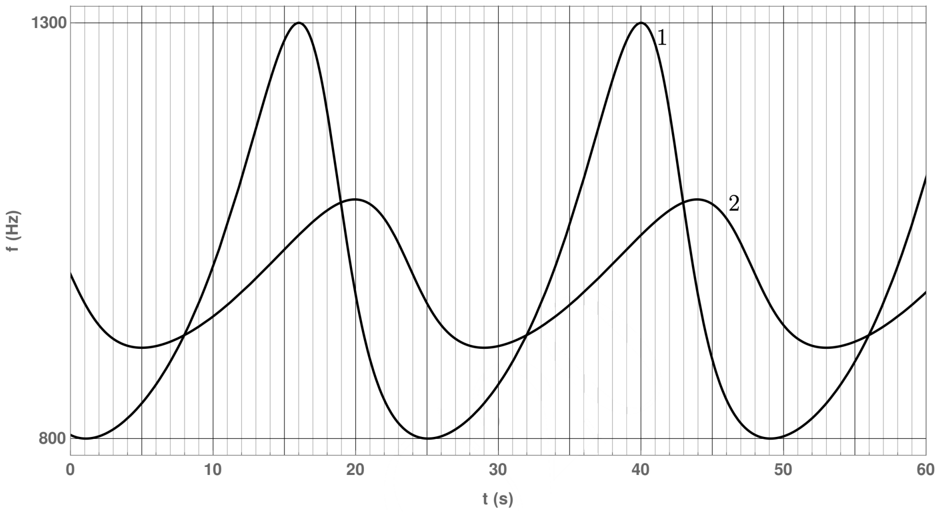
(ii) Derive the energy density $u = u(\varepsilon, E)$ in a uniformly stretched elastic band in terms of the strain ε and the Elastic (Young's) modulus E.

(iii) Use dimensional analysis to estimate the frequency-dependent elastic modulus of spacetime $E(f)$ in terms of the universal gravitational constant G, speed of light c and gravitational wave frequency f.

(iv) Estimate the maximum distance $z = z(\varepsilon, f)$ from the Earth to the gravitational wave source as a function of the strain ε and frequency f. Use the models derived previously in this problem.

1.4 Moving Source

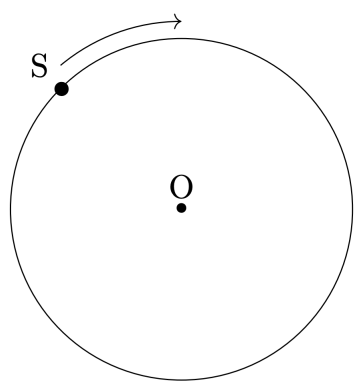
A sound source S is performing uniform circular motion with time period T. It is continuouslyemitting sound of a fixed frequency f_0. Two detectors 1 and 2 are placed somewhere in the same plane as the circular trajectory of the source. The frequency f, of the sound received by the two detectors is plotted as a function of time t as shown below (the clocks of the two detectors are synchronized). (InPhO 2020, P4)



Take the speed of the sound in the medium to be 330 m/s.

(i) Determine the time period T of the source.

(ii) The figure below shows the circular trajectory of the source S. Qualitatively mark the positions of both the detectors by indicating 1 and 2. Here O denotes the center of the trajectory. You must provide detailed justification of your answer.



(iii) Obtain the frequency f_0 of the source.

(iv) Calculate the distance D between the detectors.

1.5 Physics of Rod Instruments

Some instruments, such as xylophones and marimbas, are made with rigid rods instead of strings. The equation that describes transverse vibrations is instead

$$\frac{\partial^2 y}{\partial t^2} = -A\frac{\partial^4 y}{\partial x^4}$$

for a constant A that depends on the material and cross-sectional area. (Kevin Zhou, Waves I, P12)

(i) For a xylophone bar of length L, find the standing wave solutions and their angular frequencies. For simplicity, pretend that the solutions are sinusoidal in space, and that the bar has free ends just like a string, even though this is not true in reality.

(ii) When the bar in part (i) is hit, a certain note is sounded. What is the length of the bar that makes a note one octave higher?

(iii) The actual boundary conditions for a free bar are

$$\frac{\partial^2 y}{\partial x^2} = \frac{\partial^3 y}{\partial x^3} = 0$$

at the endpoints, and the solutions aren't purely sinusoidal in space. Compute the lowest few standing wave angular frequencies and compare them to those you found in part (i). You'll have to use a calculator or computer to do this.

(iv) A guitar or piano string satisfies the wave equation with a small additional fourth-order term

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2} - A\frac{\partial^4 y}{\partial x^4}$$

Show that the standing wave frequencies are not linearly spaced, as they would be for an ideal string, but instead are slightly more spaced out. This effect is called inharmonicity. (Hint: The spatial profiles of the standing waves are still sinusoidal.)

1.6 Physics of Drums

The top of a drum is like a string, in that it has a uniform surface mass density σ and surface tension γ. (Kevin Zhou, Waves I, P15)

(i) Waves on the drum can be described by its height $z(x, y, t)$. Find the wave equation for a drum. What is the speed of traveling waves?

(ii) Consider a square drum of side length L, where the boundaries are fixed to $z = 0$. Find the standing wave solutions and the corresponding ω. What's the lowest standing wave angular frequency?

The frequencies will not be multiples of a fundamental frequency, so they are called overtones, rather than harmonics; that's why drums don't sound like they're playing notes. (Special examples, such as the timpani, are designed to mostly excite the harmonic frequencies.)

(iii) Why does a drum sound different if you hit it near the edge, versus at the center?

1.7 More on Instruments

(i) A piano makes sound by quickly striking a string with a hammer. The seventh harmonic doesn't fit in with the rest that well. If you want to eliminate the seventh harmonic, at what point(s) can you put the hammer?

(ii) A violinist can make the note from an open string sound an octave higher by lightly touching it at a point while bowing it somewhere else. Which point(s) should be touched?

(iii) Suppose a string has its ends attached to walls. A person can set up a standing wave by holding the string at some point and moving it side to side, sinusoidally with fixed amplitude. At which point(s) should the string be driven to maximize the amplitude of a given standing wave? Assume the string experiences very little damping.

(Kevin Zhou, Waves I, P13)